

Problem 1 (17CMIMCA7)

We let $s = a + b + c$. We have,

$$9 = \sum \frac{a}{s-a} \Leftrightarrow 12 = \sum \frac{s}{s-a} \Leftrightarrow 12s = \sum \frac{s^2}{s-a};$$

$$32 = \sum \frac{a^2}{s-a} \Leftrightarrow 32 = \sum \left(\frac{s^2}{s-a} - (s+a) \right) \Leftrightarrow 32 + 4s = \sum \frac{s^2}{s-a};$$

Therefore, $12s = 32 + 4s \Leftrightarrow s = 4$. We rewrite the first two equations to get,

$$3 = \sum \frac{1}{4-a}$$

Finally,

$$122 = \sum \frac{a^3}{4-a} \Leftrightarrow 122 = \sum \frac{64}{4-a} - (16 + 4a + a^2)$$

$$\Leftrightarrow 186 + (a^2 + b^2 + c^2) = \sum \frac{64}{4-a}$$

$$\Leftrightarrow a^2 + b^2 + c^2 = 6$$

Therefore, we get $ab + bc + ca = 5$ from square $s = 4$ and subtracting the above equation. We now revisit,

$$3 = \frac{1}{4-a} + \frac{1}{4-b} + \frac{1}{4-c}$$

$$\Leftrightarrow 3(4-a)(4-b)(4-c) = \sum_{cyc} (4-b)(4-c)$$

$$\Leftrightarrow 192 - 48(a+b+c) + 12(ab+bc+ca) - 3abc = 48 - 8(a+b+c) + (ab+bc+ca)$$

$$\Leftrightarrow 192 - 48 \cdot 4 + 12 \cdot 5 - 3abc = 48 - 8 \cdot 4 + 5$$

$$\Leftrightarrow abc = 13$$

Problem 3 (Willans)

$$\left| \frac{n}{2} + \frac{1}{2} \sum_{k=0}^{n-1} \cos \left(\frac{k! - k}{k+1} \pi \right) \right| = \left| \frac{n}{2} - \frac{1}{2} \sum_{k=1}^n \cos \left(\frac{(k-1)! + 1}{k} \pi \right) \right|$$

$$= \left| \frac{n}{2} - \left(\frac{1}{2} \sum_{\text{prime } 1 \leq k \leq n} \cos \left(\frac{(k-1)! + 1}{k} \pi \right) \right) - \frac{1}{2} \sum_{\text{not prime } 1 \leq k \leq n} \cos \left(\frac{(k-1)! + 1}{k} \pi \right) \right|$$

$$= \left| \frac{1}{2} \left(\sum_{\text{prime } 1 \leq k \leq n} \cos \left(\frac{(k-1)! + 1}{k} \pi \right) - 1 \right) - \frac{1}{2} \left(\sum_{\text{not prime } 1 \leq k \leq n} \cos \left(\frac{(k-1)! + 1}{k} \pi \right) - 1 \right) \right|$$

Note that, if k is prime, $(k-1)! + 1 \equiv 0 \pmod{k}$. Additionally, $\frac{(k-1)! + 1}{k}$ is also odd for $k = 2$ and $k > 2$.

Therefore, $\cos \left(\frac{(k-1)! + 1}{k} \pi \right) = -1$ for prime k .

$$= \left| (\# \text{ primes at most } n) - \frac{1}{2} \left(\sum_{\text{not prime } 1 \leq k \leq n} \cos \left(\frac{(k-1)! + 1}{k} \pi \right) - 1 \right) \right|$$

When $k = 1$, $\sum_{\text{not prime } 1 \leq k \leq n} \cos \left(\frac{(k-1)! + 1}{k} \pi \right) - 1 = 0$.

$$= \left| (\# \text{ primes at most } n) - \frac{1}{2} \left(\sum_{\text{composite } k \leq n} \cos \left(\frac{(k-1)! + 1}{k} \pi \right) - 1 \right) \right|$$

Claim: For composite k , $\cos \left(\frac{(k-1)! + 1}{k} \pi \right) = \cos \left(\frac{1}{k} \pi \right)$.

The special case is $k = 4$ in which case $\cos \left(\frac{7}{4} \pi \right) = \cos \left(\frac{1}{4} \pi \right)$.

For $k = 6, 8$, $(k-1)! \equiv 0 \pmod{2k}$ can be checked to be true.

For $k > 8$, the smallest prime factor p of k satisfies $p \leq \sqrt{k} < \frac{k}{3}$ so $1 \leq p, 2p, \frac{n}{p} \leq n-1$. If $\frac{n}{p} = p$, then $2p \left(\frac{p}{k} \right) \mid (k-1)!$ as desired. If $\frac{n}{p} = 2p$, then $n = 2p^2$. We have $p > 2$ in which case $2(p)(2p) \mid (k-1)!$ as desired.

Therefore, we show that,

$$\pi(n) = \left| (\# \text{ primes at most } n) - \frac{1}{2} \left(\sum_{\text{composite } k \leq n} \cos \left(\frac{1}{k} \pi \right) - 1 \right) \right|$$

To finish, we show that, for all n ,

$$\begin{aligned} \sum_{\text{composite } k \leq n} \cos \left(\frac{1}{k} \pi \right) - 1 &\geq -2 \\ \Leftrightarrow \sum_{k=4}^{\infty} \cos \left(\frac{1}{k} \pi \right) - 1 &\geq -2 \\ \Leftrightarrow \sum_{k=4}^{\infty} -2 \sin^2 \left(\frac{1}{2k} \pi \right) &\geq -2 \\ \Leftrightarrow \sum_{k=4}^{\infty} \sin^2 \left(\frac{1}{2k} \pi \right) &\leq 1 \end{aligned}$$

Note that $\sin\left(\frac{1}{2k}\pi\right) \leq \frac{\pi}{2k}$ since $\frac{1}{2k}\pi > 0$.

$$\begin{aligned} &\Leftrightarrow \sum_{k=4}^{\infty} \left(\frac{\pi}{2k}\right)^2 \leq 1 \\ &\Leftrightarrow \frac{\pi^2}{4} \left(\frac{\pi^2}{6} - 1 - \frac{1}{4} - \frac{1}{9}\right) \leq 1 \end{aligned}$$

Note that $\pi^2 < (3.15)^2 < 10$.

$$\Leftrightarrow \frac{10}{4} \left(\frac{10}{6} - 1 - \frac{1}{4} - \frac{1}{9}\right) \leq 1$$

This is true.

6 clubs; 1.5 hours

Problem 6 (06SLA2)

We use induct on n .

Base case: $a_0 = 1$; $a_1 = \frac{1}{2}$. Both are above zero.

Inductive step:

$$\begin{aligned} a_n + \frac{a_{n-1}}{2} + \cdots + \frac{a_0}{n} &= 0; \\ a_{n+1} + \frac{a_n}{2} + \cdots + \frac{a_0}{n+1} &= 0 \end{aligned}$$

Therefore,

$$\begin{aligned} &\left(a_{n+1} + \frac{a_n}{2} + \cdots + \frac{a_0}{n+1}\right) - \frac{n}{n+1} \left(a_n + \frac{a_{n-1}}{2} + \cdots + \frac{a_0}{n}\right) = 0 \\ &\Leftrightarrow a_{n+1} = \left(\frac{n}{n+1} - \frac{1}{2}\right)a_n + \left(\frac{n}{n+1} \cdot \frac{1}{2} - \frac{1}{3}\right)a_3 + \cdots + \left(\frac{n}{n+1} \cdot \frac{1}{n} - \frac{1}{n+1}\right)a_0 \\ &\Leftrightarrow a_{n+1} = \left(\frac{n}{n+1} - \frac{1}{2}\right)a_n + \left(\frac{n}{n+1} \cdot \frac{1}{2} - \frac{1}{3}\right)a_3 + \cdots + \left(\frac{n}{n+1} \cdot \frac{1}{n-1} - \frac{1}{n+1}\right)a_1 \end{aligned}$$

Note that $\frac{n}{n+1} \cdot \frac{1}{k} - \frac{1}{k+1} > 0$ for $k < n$ meaning $a_{n+1} > 0$.

Problem 7 (20EGMO2)

Note that,

$$\sum_{i=1}^{2020} ((x_i + 1)(y_i + 1))^2 = 8 \sum_{i=1}^{2020} x_i^3$$

$$\Leftrightarrow \sum_{i=1}^{2020} ((x_i + 1)(y_i + 1))^2 = 4 \sum_{i=1}^{2020} x_i^3 + 4 \sum_{i=1}^{2020} y_i^3$$

However, for all i , we show that, for $|x_i - y_i| \leq 1$,

$$((x_i + 1)(y_i + 1))^2 \geq 4(x_i^3 + y_i^3)$$

With equality occurring only if $(x_i, y_i) = (1, 2), (2, 1), (0, 1), (1, 0)$. For $x_i = y_i$,

$$(x_i + 1)^4 > 8x_i^3$$

We may check that this is true for $x_i = 0, 1, 2, 3$. For $x_i \geq 4$, $(x_i + 1)^4 > x_i^4 + 4x_i^3$ is greater than $8x_i^3$.

For $x_i \neq y_i$, WLOG let $x_i < y_i$ meaning $x_i = y_i - 1$. We'd check,

$$y_i^2(y_i + 1)^2 \geq 4(y_i^3 + (y_i - 1)^3)$$

Quick check gives this is true and equality holds for $y_i = 1, 2$ only.

In conclusion, the only solutions are $a_1 = \dots = a_{1010} = 1; a_{1011} = \dots = a_{2020} = 2$ or the same thing with 0, 1 respectively.

14 clubs; 5 hours

Problem 9 (23ROU101)

We claim the solutions are $x = 0, x = 1$, which clearly work.

Rearranging the equation, we get,

$$(5^x + 5^x) + (6^x + 6^x) = (3^x + 7^x) + (3^x + 9^x)$$

To prove nothing else works, we casework,

Case 1: $x > 1$.

We show that,

$$3^x + 7^x > 5^x + 5^x$$

$$\Leftrightarrow \frac{3^x + 7^x}{2} > \left(\frac{3+7}{2}\right)^x$$

Since $x > 0$,

$$\Leftrightarrow \sqrt[x]{\frac{3^x + 7^x}{2}} > \frac{3+7}{2}$$

This is true using Power-Mean since $3^x \neq 7^x$.

Case 2: $x < 0$.

Once again,

$$\begin{aligned} 3^x + 7^x &> 5^x + 5^x \\ \Leftrightarrow \frac{3^x + 7^x}{2} &> \left(\frac{3+7}{2}\right)^x \end{aligned}$$

Since $x < 0$,

$$\Leftrightarrow \sqrt[x]{\frac{3^x + 7^x}{2}} < \frac{3+7}{2}$$

This is also true by Power-Mean and noting $3^x \neq 7^x$.

Case 3: $0 < x < 1$.

$$\begin{aligned} 3^x + 7^x &< 5^x + 5^x \\ \Leftrightarrow \frac{3^x + 7^x}{2} &< \left(\frac{3+7}{2}\right)^x \\ \Leftrightarrow \sqrt[x]{\frac{3^x + 7^x}{2}} &< \frac{3+7}{2} \end{aligned}$$

This is also true by Power-Mean and noting $3^x \neq 7^x$.

Doing the same on $3^x + 9^x = 6^x + 6^x$, we get the same results. Therefore, overall, $(5^x + 5^x) + (6^x + 6^x) < (3^x + 7^x) + (3^x + 9^x)$ for $x > 1$ and $x < 0$ and $(5^x + 5^x) + (6^x + 6^x) > (3^x + 7^x) + (3^x + 9^x)$ for $x \in (0, 1)$ as desired.

Problem 10 (04SLA2)

The sequence is unbounded.

Note that all the terms must be non-negative since $|a_{n+1} - a_{n+2}|$ is non-negative.

If $a_1 > a_2$, then $a_3 = a_2 + a_1$ for it to be positive meaning $a_3 > a_2$. Otherwise, $a_1 > a_2$.

Claim: Given a pair $a_{x+1} > a_x$ in the sequence, we may find a pair a_y, a_{y+1} later in the sequence such that $a_{y+1} - a_{x+1} \geq \min(a_{x+1} - a_x, a_x)$ and $a_{y+1} - a_y \geq a_x$.

Proof: We have $a_{x+2} = a_{x+1} \pm a_x$.

If $a_{x+2} = a_{x+1} + a_x$, then $y = x + 1$ satisfies the claim since $a_{y+1} - a_y = a_{y+1} - a_{x+1} = a_{x+2} - a_{x+1} = a_x$. If $a_{x+2} = a_{x+1} - a_x < a_{x+1}$, we cannot have $a_{x+3} = a_{x+1} - a_{x+2}$ meaning $a_{x+3} = a_{x+1} +$

$a_{x+2} = 2a_{x+1} - a_x$. In this case, $y = x + 2$ satisfies the claim since $a_{y+1} - a_{x+1} = a_{x+1} - a_x$ and $a_{y+1} - a_y = a_{x+1} > a_x$ as desired. $\square$

Applying the claim another time on a_y, a_{y+1} , we may find a z in the sequence such that $a_{z+1} - a_{y+1} \geq \min(a_{y+1} - a_y, a_y) \geq a_x$ and $a_{z+1} - a_z \geq a_y \geq a_x$. If we continue doing this, we will find more elements in the sequence after a_{z+1} that are all at least a_x greater than the previous. Taking $a_x = a_2$ or $a_x = a_1$, we may fix a_x and see that the sequence is unbounded.

20 clubs; 6.5 hours

Problem 11 (15SLA1)

We have,

$$\begin{aligned} a_{k+1} &\geq \frac{ka_k}{a_k^2 + (k-1)} \\ \Leftrightarrow \frac{1}{a_{k+1}} &\leq \frac{a_k^2 + (k-1)}{ka_k} \\ \Leftrightarrow \frac{k}{a_{k+1}} &\leq a_k + \frac{k-1}{a_k} \\ \Leftrightarrow \frac{k}{a_{k+1}} - \frac{k-1}{a_k} &\leq a_k \end{aligned}$$

Summing this,

$$\frac{k}{a_{k+1}} \leq a_1 + a_2 + \cdots + a_k$$

We now induct.

Base case: $n = 2$.

$$\frac{1}{a_2} \leq a_1$$

Therefore, $a_1 + a_2 \geq 2$.

Inductive step:

$$\Leftrightarrow a_{k+1} \geq \frac{k}{a_1 + a_2 + \cdots + a_k}$$

Therefore,

$$\Leftrightarrow a_1 + a_2 + \cdots + a_{k+1} \geq (a_1 + a_2 + \cdots + a_k) + \frac{k}{a_1 + a_2 + \cdots + a_k}$$

For $a_1 + a_2 + \cdots + a_k \geq k$, we get $a_1 + a_2 + \cdots + a_{k+1} \geq k + 1$ as desired.

Problem 12 (20USATST1)

We claim that any $r \in \left[0, \frac{1}{2}\right]$ works.

For the construction, we may first let $b_i = \frac{i(i+1)}{2}$ for all $i < N$. This works since $b_1 = 1$ works and,

$$\begin{aligned} \frac{b_{n-1}}{(n-1)^2} &> \frac{b_n}{n^2} \\ \Leftrightarrow \frac{\frac{(n-1)n}{2}}{(n-1)^2} &> \frac{\frac{n(n+1)}{2}}{n^2} \\ \Leftrightarrow \frac{n}{n-1} &> \frac{n+1}{n} \\ \Leftrightarrow n^2 &> n^2 - 1 \end{aligned}$$

This is true. Therefore,

$$\lim_{n \rightarrow \infty} \frac{b_n}{n^2} = \frac{1}{2}$$

We now pick an N . There exists some integer k such that $\left| \frac{k}{N^2} - r \right| < \frac{1}{N^2}$. Since $r \leq \frac{1}{2}$, we may ensure $k < \frac{N(N+1)}{2}$.

Afterwards, we maximize $b_{N+1}, b_{N+2}, \dots$. Note that,

$$\begin{aligned} \frac{b_n}{n^2} &> \frac{b_{n+1}}{(n+1)^2} \\ \Leftrightarrow b_{n+1} &< \left\lfloor \frac{b_n(n+1)^2}{n^2} \right\rfloor \end{aligned}$$

If we were to maximize b_{n+1} , we may ensure,

$$b_{n+1} \geq \frac{b_n(n+1)^2}{n^2} - 2$$

Therefore, we may achieve,

$$\frac{b_{n+1}}{(n+1)^2} \geq \frac{b_n}{n^2} - \frac{2}{n^2}$$

We'd have,

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{b_N}{N^2} - \frac{b_n}{n^2} &< \frac{1}{N^2} + \frac{1}{(N+1)^2} + \dots \\ &< \left(\frac{1}{N^2} + \dots + \frac{1}{(2N-1)^2} \right) + \left(\frac{1}{(2N)^2} + \dots + \frac{1}{(4N-1)^2} \right) + \dots \end{aligned}$$

$$\begin{aligned}
&< \left(\frac{1}{N^2} + \dots + \frac{1}{N^2} \right) + \left(\frac{1}{(2N)^2} + \dots + \frac{1}{(2N)^2} \right) + \dots \\
&= \frac{1}{N} + \frac{1}{2N} + \dots \\
&= \frac{2}{N}
\end{aligned}$$

The difference between $\lim_{n \rightarrow \infty} \frac{b_N}{N^2}$ and r would ultimately be less than $\frac{2}{N} + \frac{1}{N^2}$. Taking N large, we may get arbitrarily close to r as desired. [Is the same as the bound is achievable if N may need to approach infinity for this to get to r ? If so, will this lose points?]

28 clubs; 9 hours

Problem 13 (11MEX3)

The answer is two. Either $(-1, -1, \dots, -1)$ or $(1, 1, \dots, 1)$ which both clearly work.

Let $f(x) = x^2 + x - 1$. We see that $a_{k+1} = f(a_k)$. We count how many x there are such that we have $f^n(x) = x$. Each of these would correspond to exactly one valid sequence.

We induct to see that $x = -1, 1$ are the only solutions.

Case 1: $x > 1$.

We have $f(x) > x$ for

Case 2: $0 \leq x < 1$.

On this range, $f(x) < x$. Continuing applying x , eventually $f^{(k)}(x) < 0$ for some k . Since $f(x) \geq -1.25$ for all x , and we must have $x < -\frac{1+\sqrt{5}}{2} < -1.25$ for $f(x) > 0$ on $x < 0$, we may induct to see that x will remain negative after k . Therefore, it is impossible to get back to $f^{(n)}(x) = x \geq 0$.

Case 3: $-1.25 \leq x < 0, x \neq -1$

On this range, if $x < -1$, then $f(x) > -1$. If $x > -1$, then $f(x) < -1$. Therefore, n is even. We add,

$$a_2 = a_1^2 + a_1 - 1$$

$$a_3 = a_2^2 + a_2 - 1$$

...

$$a_1 = a_n^2 + a_n - 1$$

We get that,

$$n = a_1^2 + a_2^2 + \dots + a_k^2$$

However, on this range,

$$\begin{aligned}
x^2 + f(x)^2 &= x^4 + 2x^3 - 2x + 1 \\
&= (x+1)^3(x-1) + 2
\end{aligned}$$

This would be negative. If we pair up $a_1^2, a_2^2; a_3^2, a_4^2; \dots$, we find $n < a_1^2 + a_2^2 + \dots + a_k^2$ as desired.

Case 4: $x < -1.25$

$f(x) \geq -1.25$ for all x so $f^{(n)}(x) = x$ is not happening.

Problem 14 (82IMO1)

Note that,

$$\begin{aligned}
f(m+n) - f(m) - f(n) &\in \{0, 1\} \\
\Rightarrow f(m) + f(n) &\leq f(m+n) \leq f(m) + f(n) + 1
\end{aligned}$$

Immediately, we have $f(1) = 0$ since if $f(1) \geq 1$ then $f(2) \geq 2f(1) > 0$.

Claim 1: $f(3) = 1$.

Proof: For the sake of contradiction, let $f(3) \geq 2$. We'd have,

$$f(6) \geq f(3) + f(3) \geq 4$$

However,

$$\begin{aligned}
f(2) &\leq f(2) + f(2) + 1 = 1 \\
\Rightarrow f(4) &\leq 2 \Rightarrow f(6) \leq 3 \quad (!) \quad \square
\end{aligned}$$

We have $f(3) = 1$. We may similarly induct to find, $f(6) \leq 2; f(9) \leq 3; \dots$ with equality holding for $f(3n) = n$ only if $f(3k) = k$ for all $k < n$. Therefore, $f(9999) = 3333$ means $f(3k) = k$ for all positive integers $k \leq 3333$.

Claim 2: $f(6k-2) = 2k-1$ for all $k \leq 833$.

Proof: We have $f(6k-2) \leq f(6k) \leq f(2) + f(6k-2) + 1$ meaning $f(6k-2) \in \{2k, 2k-1\}$. However, if $f(6k-2) = 2k$, $f(12k-4) = 4k$ even though $f(12k-6) = 4k-2$ (we can note that $12k-6 \leq 9990$). In that case, $f(12k-4) - f(2) - f(12k-6) = 2$ which is not allowed.

Claim 3: $f(6k+2) = 2k$ for all $k \leq 330$.

Proof: $f(6k+2) \in \{f(6k) + f(2) + 1, f(6k) + f(2)\} \Leftrightarrow f(6k+2) \in \{2k, 2k+1\}$. Note that if $f(6k+2) = 2k+1$, then $f(12k+4) \geq 4k+2$. This contradicts $f(12k+4) = 4k+1$.

Using claim 3, we have $f(1982) = 660$.

34 clubs; 11 hours

Problem 15 (18SLN3)

We let n be writeable if we can write it as a sum of at least two distinct elements of a .

Lemma: If 2^x is writeable if and only if 2^{4x-6} is writeable.

Proof: To write 2^n , we may never use $a_{n+1}, a_{n+2}, \dots$ by bounding.

For 2^n to be writeable using a subset of $a_0, \dots, a_{n-1}$, we first select all $a_0, \dots, a_{n-1}$ and choose which to deselect instead. Note that,

$$\begin{aligned} \sum_{i=0}^{n-1} a_i &= \begin{cases} (2^n - 1) + 2 \left(2^{\frac{n}{2}} - 1 \right) & \text{if } n \text{ is even} \\ (2^n - 1) + 2^{\frac{n-1}{2}} + 2 \left(2^{\frac{n-1}{2}} - 1 \right) & \text{if } n \text{ is odd} \end{cases} \\ &= \left(2^n + 2^{\lfloor \frac{n}{2} \rfloor} \right) + \left(2^{\lfloor \frac{n+1}{2} \rfloor} - 3 \right) \end{aligned}$$

Therefore, 2^n is writeable if and only if $2^{\lfloor \frac{n+1}{2} \rfloor} + 2^{\lfloor \frac{n}{2} \rfloor} - 3$ is writeable.

Plugging in $n = 4k - 6$, 2^n is writeable if and only if $2^{\lfloor \frac{n+1}{2} \rfloor} + 2^{\lfloor \frac{n}{2} \rfloor} - 3 = 2^{2k-3} + 2^{2k-3} - 3 = 2^{2k-2} - 3$ is writeable. We do the same thing again with $a_0, \dots, a_{2k-3}$ to get,

$$\sum_{i=0}^{2k-3} a_i = \left(2^{2k-2} + 2^{\lfloor \frac{2k-2}{2} \rfloor} \right) + \left(2^{\lfloor \frac{2k-1}{2} \rfloor} - 3 \right) = 2^{2k-2} + 2^k - 3$$

Therefore, 2^n is writeable if and only if 2^k is writeable as desired.

Claim 1: 2^{12} is not writeable.

We found that 2^{12} is not writeable if and only if $2^{\lfloor \frac{12+1}{2} \rfloor} + 2^{\lfloor \frac{12}{2} \rfloor} - 3 = 2^7 - 3$ is not writeable.

Furthermore, $2^7 - 3$ is not writeable if and only if $2^{\lfloor \frac{7+1}{2} \rfloor} + 2^{\lfloor \frac{7}{2} \rfloor} = 2^3 + 2^4 = 24$ is not writeable. Indeed, we may check that using the elements of a at most 24: $a_0 = 1; a_1 = 3; a_2 = 6; a_3 = 10; a_4 = 20$, it is evidently impossible to make 24.

Claim 2: There are infinitely many a_n that are writeable.

Proof: Note that $2^3 = a_0 + a_2$ is writeable. This implies $2^{4 \cdot 3 - 6} = 2^6$ is writeable, which then implies $2^{4 \cdot 6 - 6} = 2^{18}$ is writeable, ... Clearly, there are infinitely many k such that 2^{2k} is writeable. When writing 2^{2k} , by bounding, we only used elements of $a_0, a_1, \dots, a_{2k-1}$. Taking,

$$2^{2k} + a_{2k} = 2^{2k+1} + 2^k = a_{2k+1}$$

This gives us a writeable a_n . Since there are infinitely many valid k , we are done. $\square$

Claim 3: There are infinitely many a_n that are not writeable.

2^{12} not writeable $\Rightarrow 2^{42}$ not writeable $\Rightarrow \dots$

As before, there are infinitely many k such that 2^{2k} is not writeable. For all such k , FTSOC let a_{2k+1} be writeable. Since $\sum_{i=0}^{2k-1} a_i = 2^{2k} + 2^{k+1} - 3 < a_{2k+1}$, we must select a_{2k} to be part of its sum by bounding. The sum of the remaining elements must be $a_{2k+1} - a_{2k} = 2^{2k}$ which is not writeable. $\square$

Problem 16 (18PTNMB4)

Note that if $|a| > 1$, then the absolute value of the terms would increase since,

$$\begin{aligned} |x_{n+1}| &\in \{2|x_n x_{n-1}| - |x_{n-2}|, 2|x_n x_{n-1}| + |x_{n-2}|\} \\ &\Rightarrow |x_{n+1}| \geq 2|x_n x_{n-1}| - |x_{n-2}| \end{aligned}$$

Since $|a| > 1$, we may induct to find that this is strictly greater than $|x_n|$. Therefore, in this case, a_n never repeats itself.

If $|a| \leq 1$, we may substitute $a = \cos(\theta)$ for some θ .

Sketch: Using induction, we may prove $x_n = \cos(F_n \theta)$ where F_n is Fibonacci.

There exists some $x_n = 0$ if and only if $F_n \theta \equiv \frac{\pi}{2} \pmod{\pi}$. Since F_n is an integer, $\theta = \frac{p}{2q} \pi$ for some integers p, q (where p may be even if necessary).

Sketch: There's a finite number of ordered pairs $(F_n, F_{n+1}) \pmod{2q}$ meaning the sequence will repeat.

42 clubs; 15 hours

Problem 17 (21AMO5)

Here's my solution sketch (I'm posting here because I'm unsure whether this works). I have the full solution written out too if anyone wants it.

Sketch:

Claim: $a_n \leq 1$ for all odd n .

FTSOC, if, WLOG, $a_1 > 1$. We WLOG take $a_2 \leq a_{2n}$ and get $a_2 \leq \frac{2}{a_1}$. We then get $a_3 = a_2 + a_1 \leq \frac{2}{a_1} + a_1 < \frac{1}{a_1}$.

Continuing, we'd get $a_5 > a_1$, so if n is even we'd induct to get $a_1 < a_5 < a_9 < \dots < a_{2n+1} = a_1$ (with indices taken mod $2n$). However, if n is odd,

$$a_1 > 1 \Rightarrow a_3 < 1 \Rightarrow a_5 > 1 \dots a_{2n+1} = a_1 < 1$$

Claim: $a_n \geq 1$ for all odd n .

If not then $a_{n+2} > 1$. (!)

Either are impossible so $a_{\text{odd}} \equiv 1$ and $a_{\text{even}} \equiv 2$.

Problem 18 (18IMO2)

We claim the answer is all multiples of 3. This is attainable with the sequence $0, -1, 1, 0, -1, 1, \dots$. To prove this is the only solution, we show $a_i = a_{i+3}$ for all i .

$$a_i a_{i+1} + 1 = a_{i+2} \Leftrightarrow a_{i-1} a_i a_{i+1} = a_{i+2} a_{i-1} - a_{i-1}$$

On the other hand,

$$a_{i-1} a_i + 1 = a_{i+1} \Leftrightarrow a_{i-1} a_i a_{i+1} = a_{i+1}^2 - a_{i+1}$$

We have $a_{i+2} a_{i-1} - a_{i-1} = a_{i+1}^2 - a_{i+1}$. Summing this across $i = 1, \dots, n$,

$$\begin{aligned} \sum a_i a_{i+3} - a_{i-1} &= \sum a_{i+1}^2 - a_{i+1} \\ \Leftrightarrow \sum a_i a_{i+3} - a_i &= \sum a_i^2 - a_i \\ \Leftrightarrow \sum a_i a_{i+3} &= \sum a_i^2 \end{aligned}$$

By AM-GM, we require $a_i = a_{i+3}$ for all i as desired.

52 clubs; 16 hours

Problem 20 (08RUS104)

Note that,

$$\begin{aligned} a_{n+1} &= \frac{1 + a_n + a_n b_n}{b_n} \\ \Leftrightarrow a_{n+1} + 1 &= \frac{(a_n + 1)(b_n + 1)}{b_n} \end{aligned}$$

We plug in $A_n = a_n + 1; B_n = b_n + 1$. By symmetry,

$$A_{n+1} = \frac{A_n B_n}{B_n - 1}; B_{n+1} = \frac{A_n B_n}{A_n - 1}$$

Note that,

$$\frac{1}{A_{n+1}} - \frac{1}{B_{n+1}} = \frac{B_n - A_n}{A_n B_n}$$

$$\Leftrightarrow \frac{1}{A_{n+1}} - \frac{1}{B_{n+1}} = \frac{1}{A_n} - \frac{1}{B_n}$$

Note that $\frac{1}{A_1} - \frac{1}{B_1} = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$. FTSOC if $A_n \geq 6$ for some n , then B_n either does not exist or is negative, which is impossible since clearly $a_n, b_n > 0$ for all n .

Problem 21 (18SLN5)

We claim the answer is no.

FTSOC, let $a^2 = xy$; $b^2 = zt$ for some integers a, b .

$$a^2 - b^2 = x + y = z + t$$

If we write the quadratics,

$$F(X) = X^2 - (a^2 - b^2)X + a^2;$$

$$G(X) = X^2 - (a^2 - b^2)X + b^2;$$

Note that the solutions to the quadratics would be x, y and z, t respectively by Vieta's. Using the quadratic formula, the solutions are,

$$\frac{(a^2 - b^2) \pm \sqrt{(a^2 - b^2)^2 - 4a^2}}{2}$$

$$\frac{(a^2 - b^2) \pm \sqrt{(a^2 - b^2)^2 - 4b^2}}{2}$$

Note that $a^2 - b^2 = xy - zt = x + y = z + t$ must be a multiple of four if we use casework on $x, y, z, t \pmod{4}$. Therefore, $(a^2 - b^2)^2 - 4a^2, (a^2 - b^2)^2 - 4b^2$ are both perfect squares. We let $d = a - b$; $s = a + b$.

$$(a^2 - b^2)^2 - 4a^2 = d^2s^2 - (s + d)^2$$

$$(a^2 - b^2)^2 - 4b^2 = d^2s^2 - (s - d)^2$$

If these are both squares, their product is also a square.

$$(d^2s^2 - (s + d)^2)(d^2s^2 - (s - d)^2)$$

$$= (d^2s^2 - s^2 - d^2)^2 - (2sd)^2$$

We check that this quantity is not equal to $(d^2s^2 - s^2 - d^2 - 1)^2$. If it were,

$$(2sd)^2 = 2d^2s^2 - 2s^2 - 2d^2 - 1$$

$$\Rightarrow 2d^2s^2 < 0 (!)$$

This quantity is also not equal to $(d^2s^2 - s^2 - d^2 - 2)^2$ for if it were,

$$(2sd)^2 = 2(2d^2s^2 - 2s^2 - 2d^2 - 2)$$

$$\Leftrightarrow 0 = -4s^2 - 2d^4 - 4$$

Therefore,

$$(d^2s^2 - s^2 - d^2)^2 - (2sd)^2 \leq (d^2s^2 - s^2 - d^2 - 3)^2$$

$$\Leftrightarrow 3(2d^2s^2 - 2s^2 - 2d^2 - 3) \leq (2sd)^2$$

$$\Leftrightarrow 2d^2s^2 \leq 6s^2 + 6d^2 + 9$$

$$\Leftrightarrow (d^2 - 3)(s^2 - 3) \leq \frac{13}{2}$$

Note that $s > d$ and $s - d$ is even. For $s = 2, d = 4$ is the only solution. We may verify that $2^2 4^2 - (2 + 4)^2 = 28$ is not a square. For $s = 1, d^2s^2 - (s + d)^2 < 0$ which is also impossible. There are no possible cases where this work, so we are done.

62 clubs; 20 hours

I might finish P19, 22-25 later but I've had enough algmanip for this month lol